11

Links to Classical Quadrature

This section returns to the univariate integration problem. In this setting, we will draw the many connections between classic and Bayesian quadrature. We will also use the setting to illustrate some deeper points about probabilistic integration, and, more broadly, about Probabilistic Numerics.

▶ 11.1 The Bayesian Trapezoidal Rule

A particularly interesting Bayesian quadrature scheme is obtained by choosing the mean $m(x) = 0$, $\forall x$, and the covariance

$$k(x, x') = \theta^2 \big(\min(x, x') - \chi\big), \tag{11.1}$$

with constants $\theta \in \mathbb{R}_+$ and $\chi \in \mathbb{R}, \chi < a$, to set

$$p(f) = \mathcal{GP}\Big(f; 0, \theta^2 \big(\min(x, x') - \chi\big)\Big). \tag{11.2}$$

As we saw on p. 50 (see also Figure 5.3), this defines the *Wiener process* with starting time χ and intensity θ – a fundamental process, covering a particularly large hypothesis space.

So what is the integration rule arising from this prior? Assume function evaluations at an ordered set of nodes

$$X = [x_1, x_1 + \delta_1, x_1 + \delta_1 + \delta_2, \ldots, x_1 + \sum_{i=1}^{N-1} \delta_i] =: [x_1, x_2, \ldots, x_N]$$

for $\delta_i \in \mathbb{R}^+, \forall i$. To simplify the exposition, we assume $a \leq x_i \leq b \; \forall i$. The marginal prior mean $\mathfrak{m}_0$ of Eq. (10.5) vanishes because of the vanishing prior mean $m(x) = 0$ chosen above, and the integrals from Eqs. (10.6) and (10.7) evaluate to

$$\mathfrak{K} = \iint_a^b k(x, x') \, \mathrm{d}x \mathrm{d}x' = \theta^2 \left(\frac{1}{3} b^3 - a^2 b + \frac{2}{3} a^3 - \chi (b-a)^2 \right),$$

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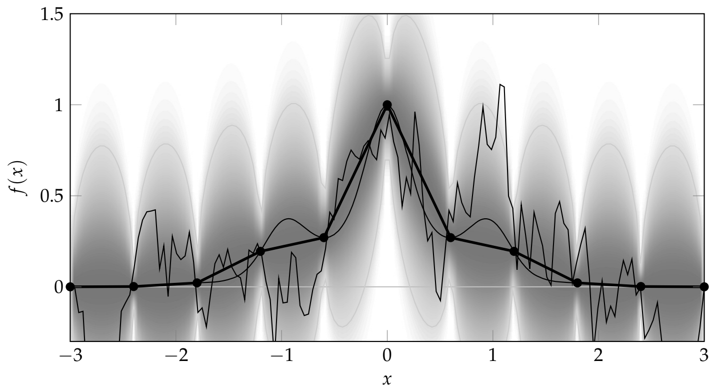

Figure 11.1: The posterior measure over f under a Wiener process prior, after 11 evaluations at equidistant points (black circles). The plot shows the piecewise linear posterior mean of this process in solid black and the probability density function as a shading. A single sample from the posterior is shown for comparison to the integrand (both thin black). The sample is clearly much less regular than the integrand.

and

$$
\begin{aligned}
\mathfrak{k}(x_i) &= \int_a^b k(x, x_i)\,\mathrm{d}x = \theta^2 \left(\int_a^{\min(b,x_i)} x\,\mathrm{d}x + \int_{\min(b,x_i)}^b x_i\,\mathrm{d}x - \chi(b-a) \right) \\
&= \theta^2 \left(\frac{1}{2}(\min(b, x_i)^2 - a^2) + x_i(b - \min(b, x_i)) - \chi(b-a) \right) \\
&= \theta^2 \left(x_i b - \frac{1}{2}(a^2 + x_i^2) - \chi(b-a) \right) \quad \text{by the assumption } a \leq x_i \leq b \text{ above.}
\end{aligned}
$$

These results provide one possible implementation, but as such they do not say much about the properties of this quadrature rule. As it turns out, they are a disguise for a well-known idea. A simple way to see this is to note that the posterior mean over f is not just a weighted sum of the observations Y (Eq. (10.8)), but also a weighted sum of kernel functions at the locations $\{x_i\}$:

$$
\mathbb{E}_{p(f|Y)}(f(x)) = k_{xX} \underbrace{k_{XX}^{-1} Y}_{=:\alpha} = \sum_i k(x, x_i)\alpha_i. \tag{11.3}
$$

The $k(x, x_i)$ of Eq. (11.1) are piecewise linear functions, each with a sole non-differentiable locus $x = x_i$. Thus the posterior mean of Eq. (11.3) is a sum of piecewise linear functions, hence itself piecewise linear, with "kinks" – points of non-differentiability – only at the input locations X; see Figure 11.1. As this is the posterior mean conditioned on Y, that piecewise linear function with N kinks has to pass through the N nodes in Y. Assuming, for simplicity, $x_1 = a, x_N = b$, there is only one such function[1] on $[a, b]$: the *linear spline* connecting the evaluations: for $a \leq x_i < x < x_{i+1} \leq b$,

$$
\mathbb{E}_{p(f|Y)}(f(x)) = f(x_i) + \frac{x - x_i}{\delta_i}(f(x_{i+1}) - f(x_i)).
$$

The expected value of the integral is the integral of the expected value,[2] written

[1] There are some technicalities to consider if x_1 does not coincide with a, because the choice of the starting time χ affects the extrapolation behaviour on the left. One resolution is to choose the asymptotic setting $\chi \to -\infty$, which gives rise to a constant extrapolation, known as the *natural spline* (Minka, 2000) (Wahba, 1990, pp. 13–14). That same solution is also found by the filter of §11.1.1, which does not require a starting time.

[2] The two integrals can be exchanged due to Fubini's theorem, which states that this is possible whenever the (here: bivariate, over x and f) integrand is absolutely integrable, which is true for integrands fulfilling the assumptions above.

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$$\begin{aligned}\mathbb{E}_{p(f|Y)}\left(\int_a^b f(x)\,\mathrm{d}x\right) &= \int_a^b \mathbb{E}\big(f(x)\big)\,\mathrm{d}x \\ &= \sum_{i=1}^{N-1} \frac{\delta_i}{2}\big(f(x_{i+1}) + f(x_i)\big).\end{aligned} \tag{11.4}$$

This is the *trapezoidal rule*.[3] It is arguably the most basic quadrature rule, second only to Riemann sums. Hence we have the following result.

[3] Davis and Rabinowitz (1984), §2.1.4

Theorem 11.1. *The trapezoidal rule is the posterior mean estimate*[4] *for the integral* $F = \int_a^b f(x)\,\mathrm{d}x$ *under any centred Wiener process prior* $p(f) = \mathcal{GP}(f; 0, k)$ *with* $k(x, x') = \theta^2\big(\min(x, x') - \chi\big)$ *for arbitrary* $\theta \in \mathbb{R}_+$ *and* $\chi < a \in \mathbb{R}$.

[4] Because the mean of a Gaussian distribution coincides with the location of maximum density, the trapezoidal rule is also the maximum a posteriori estimate associated with this setup.

This is an important result: explicitly, it states that the trapezoidal rule *is* Bayesian quadrature, for a particular choice of covariance function. This foundational algorithm of classic quadrature has a clear probabilistic numerical interpretation.

It is worth noting that the trapezoidal rule's simplicity makes it rarely the best integration rule in practice. Nonetheless, we use the simplicity of the trapezoidal rule to drive intuitions applicable to all numerical methods. For stronger methods, see §10 and §11.4.

▷ 11.1.1 Alternative Derivation as a Filter

Another way to arrive at Theorem 11.1 is to phrase the joint inference on (f, F) under the Wiener process model for the integrand as a Kalman filter (§5). Doing so provides an alternative proof of Theorem 11.1 and a convenient way to compute the posterior variance on F, which we have so far ignored in the analysis. A downside of the filtering formulation is that it does not extend to the multivariate domain: the form of Eq. (10.2) is more general. However, the filtering framework affords some generalisations, introduced below.

Further, the algorithmic implementation of the filter serves as an explicit example that a probabilistic formulation of computation need not have prohibitively high cost. In fact, the probabilistic numerical algorithm can be identical in cost to a classical form. At this point, it may be helpful to review §5.3 on *stochastic differential equations* (SDEs), from which we adopt notation.

To simplify notation, define the anti-derivative

$$F_x = \int_a^x f(\tilde{x})\,\mathrm{d}\tilde{x} \qquad \text{for } x \geq a, \tag{11.5}$$

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and consider the state-space $z(x) = \left[F_x, f_x\right]^\intercal$. This state space links the integrand and its anti-derivative, allowing the Gaussian process prior of Eq. (11.2) to equivalently be formulated as the linear SDE

$$\mathrm{d}z(x) = Fz(x)\,\mathrm{d}x + L\,\mathrm{d}\omega_t \quad \text{with}$$
$$F = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad L = \begin{bmatrix} 0 \\ \theta \end{bmatrix}, \quad \text{and } z(\chi) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

As already derived in Eqs. (5.25) and (5.26), from Eq. (5.21) we find that this SDE is associated with the discrete-time quantities (for a step of length $x_{i+1} - x_i =: \delta_i$)

$$A_i = \begin{bmatrix} 1 & \delta_i \\ 0 & 1 \end{bmatrix}, \qquad Q_i = \theta^2 \begin{bmatrix} \delta_i^3/3 & \delta_i^2/2 \\ \delta_i^2/2 & \delta_i \end{bmatrix}.$$

Using $H = \begin{bmatrix} 0 & 1 \end{bmatrix}$ and setting $R = 0$ to encode the likelihood $p(y_i \mid f) = \delta(y_i = f(x_i))$, we can thus write the steps of the Kalman filter (Algorithm 5.1) explicitly,[5] and find that they simplify considerably. The mean and covariance updates in lines[6] 7 and 8 of Algorithm 5.1 are simply

$$m_i = \begin{bmatrix} [m_{i-1}]_1 + \delta_i/2(y_i + [m_{i-1}]_2) \\ y_i \end{bmatrix}, \tag{11.6}$$
$$P_i = \begin{bmatrix} [P_{i-1}]_{11} + \delta_i^3/12 & 0 \\ 0 & 0 \end{bmatrix}. \tag{11.7}$$

[5] Since the goal is to infer the definite integral F^b at the right end of the domain, there is no need to also run the smoother (Algorithm 5.2). It could be used, however, to also construct estimates for the anti-derivative F_x (Eq. (11.5)) at arbitrary locations $a \leq x < b$.

[6] Note that the symbol z has a different meaning in this chapter than in Algorithm 5.1.

If we allow for an observation at $x_1 = a$, then the initial values of the SDE are irrelevant. Since we know $F_a = 0$ by definition (thus with vanishing uncertainty), the natural initialisation for the filter at $x_1 = a$ is

$$m_1 = \begin{bmatrix} 0 \\ f(a) \end{bmatrix}, \qquad P_1 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The filter thus takes the straightforward form of Algorithm 11.1. This algorithm is so simple that it barely makes sense to spell it out. The significance of this result is that Bayesian inference on an integral, using a nonparametric Gaussian process with N evaluations of the integrand, can be performed in $\mathcal{O}(N)$ operations.

Put another way, the simple software implementation of the trapezoidal rule is *identical* to that of a particular Bayesian quadrature algorithm.

Exercise 11.2 (easy). *Convince yourself that Eqs.* (11.6) *and* (11.7) *indeed arise as the updates in Algorithm 5.1 from the choices or A, Q, H, R made above. Then show that the resulting mean estimate m_N at $x = b$ indeed amounts to the trapezoidal rule (e.g. by a telescoping sum). That is,*

$$\mathbb{E}(F) = \sum_{i=1}^{N-1} \frac{\delta_i}{2}(f_{i+1} + f_i),$$
$$\operatorname{var}(F) = \frac{\theta^2}{12} \sum_{i=1}^{N-1} \delta_i^3.$$

In practice, the algorithm could thus be implemented in this simpler (and parallelisable) form. Note again, however, that this algorithm is not a good practical integration routine, only a didactic exercise. See §11.4 and in the literature cited above for more practical algorithms.

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```
procedure INTEGRATE(@f, a, b, N, θ)
    δ := (b − a)/(N − 1)                        // choose step size
    x ← a, y_1 = f(a), m ← 0, v ← 0,            // initialise
    for i = 2, ..., N do
        x ← x + δ                               // step
        y_i ← f(x)                              // evaluate
        m ← m + δ/2(y_{i−1} + y_i)              // update estimate
        v ← v + δ^3/12                          // update error estimate
    end for
    return E(F) = m, var(F) = θ^2 v             // probabilistic output
end procedure
```

Algorithm 11.1: The probabilistic trapezoidal rule formulated as a filter. The algorithm takes a handle to the integrand f, the integration limits, a, b, a budget of N evaluations, and an externally set scale θ^2 (see below for how to adapt this at runtime).

▷ 11.1.2 Uncertainty

Formulating univariate integration as inference – as the construction of a posterior distribution over a latent quantity, under certain prior assumptions – we have "re-discovered" an old and well-trusted method for this numerical task. Though simplistic, this is an encouraging result for Probabilistic Numerics. Evidently, probabilistic numerical algorithms do not have to be abstract and involved, but can be quite simple indeed.

But the probabilistic formulation not only yielded a new way to derive the well-known trapezoidal rule, but something new: an estimator for the *error* of the trapezoidal estimate, in the form of the posterior variance $\mathfrak{v}$ in Eq. (10.2). For the concrete choice of the Wiener process prior (11.2), we saw that this expression simply evaluates to

$$\begin{aligned}\mathfrak{v} = \operatorname{var}_{p(f\mid Y,X)}(F) &= \iint_a^b \mathbb{V}(x,x')\,\mathrm{d}x\mathrm{d}x' \\ &= \frac{\theta^2}{12}\sum_{i=1}^{N-1}\delta_i^3.\end{aligned} \tag{11.8}$$

We will have to ask whether this value – which, again, clearly does not depend on the collected values y_i at all – has any sensible interpretation as an error estimate. Before we address this issue, however, we find that its form can be used as a pleasingly simple way to fix the evaluation design, the step sizes $[\delta_i]_{i=1,\dots,N-1}$.

▷ 11.1.3 Regular Grids as the Maximally-Informative Design

In many basic implementations of the trapezoidal rule, the evaluation nodes $[x_1, \dots, x_N]$ are placed on a regular grid. The above result motivates this choice probabilistically, without appealing

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to classical analysis. As mentioned in Eq. (10.2), a natural design rule is to require $\operatorname{var}(F)$ to be minimised. It follows from Eq. (11.8) that this is achieved precisely when the evaluation points lie on an *equidistant grid*.[7] See §11.4 for more discussion on how node-placement in classic quadrature rules relates to optimal design from the probabilistic perspective.

[7] To see this, first encode the assumption (made above) that the first and last node must lie on the boundaries, by setting

$$\delta_{N-1} = (b-a) - \sum_{i=1}^{N-2} \delta_i.$$

The gradient w.r.t. δ_j, $j < N-1$ is then

$$\frac{\partial \operatorname{var}(F)}{\partial \delta_j} = \frac{\theta^2}{4}[\delta_j^2 - \delta_{N-1}^2].$$

Setting this to zero (recall that all δ_i must be positive) gives $\delta_j = \delta_{N-1}$, $\forall j \neq N-1$. Without the formal requirement of $x_1 = a, x_N = b$, it actually turns out (Sacks & Ylvisaker, 1970 & 1985) that the best design is

$$x_i = a + (a-b)\frac{2i}{2N+1}.$$

This leaves a little bit more room on the left end of the domain than on the right, due to the time-directed nature of the Wiener process prior. E.g., for $N = 2$, the optimal nodes on $[a,b] = [0,1]$ are at $[2/5, 4/5]$.

▷ 11.1.4 Adaptive Error Estimation

Just like the mean estimate of the trapezoidal rule (11.4), the equidistant grid as an optimal design choice for the rule is *independent* of the scale θ^2 of the prior. Hence, there is a family of models $\mathcal{M}(\theta)$, parametrised by $\theta \in \mathbb{R}_+$, so that every Gaussian process prior in that family gives rise to the same design (the same choice of evaluation nodes), and the same trapezoidal estimation rule. The associated estimate for the square error, the posterior variance, of these models is given by Eq. (11.8). If we choose the design with equidistant steps as derived above, that expression is given by

$$\operatorname{var}(F) = \frac{\theta^2}{12}\sum_{i=1}^{N-1}\left(\frac{b-a}{N-1}\right)^3 = \frac{\theta^2(b-a)^3}{12(N-1)^2}, \qquad (11.9)$$

which means that the standard deviation $\operatorname{std}(F) = \sqrt{\operatorname{var}(F)}$, an estimate for the absolute error, contracts at a rate $\mathcal{O}(N^{-1})$, and thus more rapidly than the $\mathcal{O}(N^{-1/2})$ of the Monte Carlo estimate.

Hence the choice of the kernel k, other than the scale θ, determines the *rate* at which the error estimate $\operatorname{var}(F)$ contracts, while θ itself provides the constant *scale* of the error estimate. This situation mirrors the separation, in classical numerical analysis between *error analysis* (rate) and *error estimation* (scale). The algebraic form of the estimation rule is difficult to fundamentally change at runtime without major computational overhead, so its properties (e.g. rate) are studied by abstract analysis. The scale, on the other hand, relates to an estimate of the concrete error of the estimate, and should be estimated at runtime.

Sections 6 and 6.3 introduced the mechanism of conjugate prior hierarchical inference on θ: using a Gamma distribution to define a prior on the inverse scale θ^{-2}, the joint posterior over θ and f, F remains tractable, and can be used to address the error estimation problem. Given the prior $p(\theta^{-2}) = \mathcal{G}(\theta^{-2}; \alpha_0, \beta_0)$, the posterior on θ^{-2} can be written using the recursive terms in the Kalman filter as (reproduced for conve-

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nience from Eq. (6.14))

$$p(\theta^{-2} \mid [Y]_{1:N}) = \mathcal{G}\left(\theta^{-2}; \alpha_0 + \frac{N}{2}, \beta_0 + \frac{1}{2}\sum_{i=1}^{N} \frac{(y_i - Hm_i^-)^2}{H\tilde{P}_i^- H^\intercal}\right)$$
$$=: \mathcal{G}(\theta^{-2}, \alpha_N, \beta_N). \tag{11.10}$$

This posterior on θ is associated with a corresponding Student-t posterior marginal on the integrand F (see Eq. (6.9)),

$$p(F \mid [Y]_{1:N}) = \int \mathcal{N}(F; \mu_{F|Y}, \theta\sigma_F^2)\mathcal{G}(\theta^{-2}, \alpha_N, \beta_N)\, \mathrm{d}\theta^{-2}$$
$$= \mathrm{St}\left(F; \mu_{F|Y}, \frac{\alpha_N}{\beta_N \sigma_F^2}, 2\alpha_N\right),$$

where $\sigma_F^2 := \mathrm{var}_{|Y}(F)/\theta^2$ is the standardised posterior variance for F under the "unit" kernel, and $\mu_{F|Y} := \mathbb{E}_{\mathcal{M}}(F \mid Y)$ is the corresponding posterior mean for F. The mean of the Student-t coincides with $\mu_{F|Y}$. Its variance is[8]

$$\frac{\beta_N}{\alpha_N - 1}\sigma_F^2 = \frac{\beta_0 + \frac{N}{2}\theta_{\mathrm{ML}}^2}{\alpha_0 + \frac{N}{2} - 1}\sigma_F^2. \tag{11.11}$$

[8] E.g. Eq. (B.66) in Bishop (2006). For large values of N, the shape of the distribution approaches that of a Gaussian, with mean $\mu_{F|Y}$ and variance given by Eq. (11.11).

From Eq. (11.10) we see that, in contrast to Eq. (11.8), this estimated variance now actually depends on the function-values collected in Y. For the specific choice of the Wiener process prior (11.2), the values collected in β_N in Eq. (11.10) become[9]

$$\begin{aligned}\beta_N &= \beta_0 + \frac{1}{2}\sum_{i=1}^{N} \frac{(y_i - Hm_i^-)^2}{H\tilde{P}_i^- H^\intercal} \\ &= \beta_0 + \frac{1}{2}\left(\frac{f(x_1)^2}{x_1 + \chi} + \sum_{i=2}^{N} \frac{\big(f(x_i) - f(x_{i-1})\big)^2}{\delta_i}\right).\end{aligned} \tag{11.12}$$

[9] If necessary, the second line of Eq. (11.12) can be used to fix χ to its most likely value, given by

$$\chi_{\mathrm{ML}} := f(x_1)^2 \left(\frac{1}{N-1}\sum_{i=2}^{N} \frac{\big(f(x_i) - f(x_{i-1})\big)^2}{\delta_i}\right)^{-1} - x_1.$$

However, if $x_1 = a$, and the first evaluation y_1 is made without observation noise, the value of χ has no effect on the estimates.

For reference, Algorithm 11.2 on p. 97 provides pseudo-code and highlights again that this Bayesian parameter adaptation can be performed in linear cost, by collecting a running sum of the local quadratic residuals $(f(x_i) - Hm_i^-)^2$ of the filter.

What is the cost of adding uncertainty to a computation? In the case of the trapezoidal rule, Algorithm 11.2 gives the answer, which depends on how one defines computation. If we count the number of evaluations of f, then probabilistic inference can be offered at *no* computational overhead over the classic equivalent. If we count the overall number of computations, then we need to consider only the additional accumulation of the terms $f(x_i) - Hm_i^-$. Hence the overhead is a small percentage of the cost of the classic method.

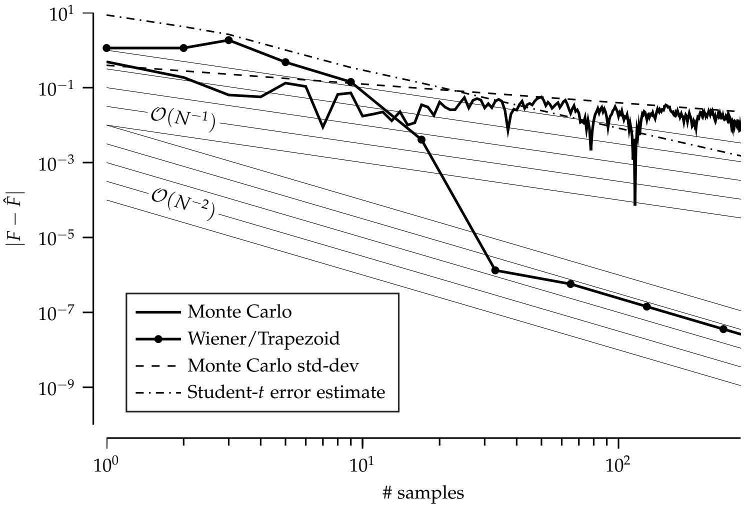

Figure 11.2: Convergence for Monte Carlo and trapezoidal rule quadrature estimates, along with different error estimates. The shown instance of Monte Carlo integration converges as $\mathcal{O}(N^{-1/2})$, as suggested by Lemma 9.2 (theoretical standard-deviation from Eq. (9.5) shown in dashed black). The trapezoidal rule overtakes the quality of the MC estimate after eight evaluations, and begins to approach its theoretical convergence rate for differentiable integrands, $\mathcal{O}(N^{-2})$ (each thin line corresponds to a different multiplicative constant). The non-adaptive GP error estimates of the form const./N (Eq. (11.9)) are under-confident. So is the adaptive Student-t error estimate (dash-dotted, as in Eq. (11.11)), reflecting the overly conservative assumption of continuity but non-differentiability in the Wiener process prior. Nevertheless, the adaptive error estimate contracts faster than the non-adaptive rate of $\mathcal{O}(N^{-1})$.

▷ 11.1.5 Convergence Rates

How good, in practice, is the probabilistic numerical trapezoidal rule? Recall that this rule was derived by assigning, ad hoc, a Wiener process prior over the integrand f and placing evaluation nodes x_i to minimise the posterior variance on the integral F: these choices are far from canonical. Figure 11.2 compares the convergence of the estimator $\mathbb{E}_{\mathcal{M}}(F)$ with that of the Monte Carlo integration estimate. To create this plot, the true integral was evaluated to high precision using another, advanced quadrature routine.

The solid black line of the Monte Carlo estimate varies stochastically, its trend described by the standard deviation $\sqrt{\operatorname{var}_p(w)/N}$ convergence of Lemma 9.2. Compared to this stochastic rate, the trapezoidal rule estimate converges much faster: after $N = 32$ evaluations of f, the absolute error between the true integral F and the estimate $\hat{F}$ of the trapezoidal rule is roughly 1.3×10^{-6}. To reach the same fidelity with the Monte Carlo estimator, the expected number of required function evaluations is $N \sim 8.8 \times 10^{10}$, or 2.75 billion times more evaluations. So using the trapezoidal rule allows a large saving of computational cost in this situation.[10]

[10] Again, the argument here is not that the trapezoidal rule is particularly efficient (it is not!), but that even the limited prior information encoded in the Wiener process allows for a drastic increase in convergence rate over the Monte Carlo rule, which can be interpreted as arising from a maximally "uninformative prior" (see §12.2).

What is the reason for this increase in performance? The probabilistic interpretation for the trapezoidal rule offers an explanation, in form of the prior in (11.2). It defines a hypothesis space which assigns non-zero probability measure to only continuous functions f. This is a more restrictive assumption than that of Monte Carlo, which only requires the integrand to

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be integrable. But of course it is a *correct* assumption, because the integrand of (9.1) is indeed continuous (even smooth).

After about $N = 64$ evaluations, the trapezoidal rule settles into a relatively homogeneous convergence at a rate of approximately $\mathcal{O}(N^{-2})$. This behaviour is predicted by classical analyses[11] of this rule for differentiable integrands like this one.[12] The probabilistically constructed non-adaptive error estimate of the trapezoidal rule (Eq. (11.8)) predicts a more conservative, slower, convergence rate $\mathcal{O}(N^{-1})$. We know from Eq. (11.9) that this is a direct consequence of the Wiener process prior assumption: draws from Wiener processes are very rough functions (almost surely continuous but not differentiable). While there is no direct classic analysis for this hypothesis class of Wiener samples itself, there is classical analysis of the Trapezoid rule for Lipschitz continuous functions that agrees with the posterior error estimate and predicts a linear error decay.[13] Even the adaptive error estimate arising from Eq. (11.11), although converging faster than the $1/N$ rate of the GP posterior standard-deviation, still yields a conservative estimate (the dash-dotted line in Figure 11.2).

The Wiener process is such a vague prior that it not only fails to capture salient analytical properties of the integrand, but also gives overly conservative estimates! This kind of insight is a strength of the probabilistic viewpoint. As we have associated the trapezoidal rule with the unnecessarily broad Wiener-process prior, that classic method, too, now seems underconstrained. Even without further analysis, we can strongly conjecture that better algorithms must be feasible for integrands like that of Eq. (9.1). And the GP prior with its defining parameters m, k provides a concrete handle guiding the search for such rules. We should seek to incorporate additional, helpful prior knowledge, to concentrate prior probability mass around the true integrand.

The space of such priors for integrands is large. It includes models that give rise to other, more advanced, classical quadrature rules (see §11.4), but also many others that are not associated with a classic method.[14] Bayesian quadrature is thus a framework for the design of customised algorithms for specific problems. The remainder of this chapter contain some examples. The following section studies one particularly straightforward way to incorporate knowledge[15] about the smoothness of the integrand.

[11] Davis and Rabinowitz (1984), p. 53

[12] The intuition for the corresponding proof is that, if the integrand is continuously differentiable, then, by the mid-point rule, the infimum and supremum of f' give an upper and lower bound on the deviation of the true integral in a segment $[x_i, x_{i+1}]$ from the integral over the linear posterior mean in that segment, and that deviation drops quadratically with the width of the segments.

[13] Davis and Rabinowitz (1984), p. 52. Note, however, that draws from the Wiener process are actually almost surely not Lipschitz. The nearest class of continuity that can be shown to contain them is that of Hölder-1/2 continuous functions; a very rough class for which the cited theorem can only guarantee $\mathcal{O}(N^{-1/2})$ convergence.

[14] Not every classical quadrature rule can be derived from a Gaussian process prior on the integrand – at least not a natural prior. An example of a method that can *not* be derived in this way is the piece-wise quadratic interpolation rule

$$\int_a^b f(x)\,\mathrm{d}x \approx \sum_i^{N-2} \frac{x_{i+1} - x_{i-1}}{2} \times \big(f(x_{i-1}) + 4f(x_i) + f(x_{i+1})\big),$$

(for a regular grid $\{x_i\}_{i=1,\dots,N}$), which, depending on national allegiance, is either called *Kepler's* or *Simpson's rule*. There is actually a "contrived" probabilistic interpretation for this rule: given a fixed evaluation grid, it arises from a Gaussian prior on the weights for *piecewise* parametric polynomial features (Diaconis, 1988). But that interpretation is so far-fetched (among other problems, it requires a constant set of pre-determined evaluation nodes) as to seem useless.

[15] There is a difference between *knowledge* and *assumptions*. A prerequisite for any numerical computation is the explicit definition of the task, in a machine-readable form like (9.1) – that is, an encoding in a formal (programming) language. Thus, properties like the existence of continuous derivatives can be truly known, not just assumed.

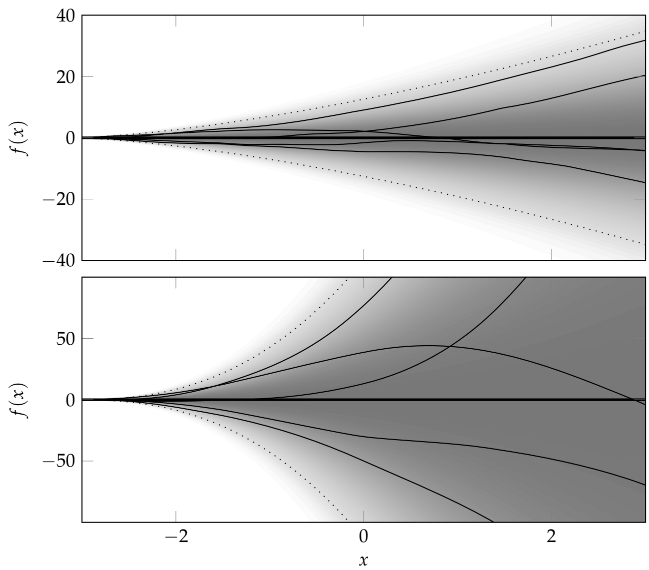

Figure 11.3: Draws from, and marginal densities of, the zero-mean integrated (top) and twice-integrated (bottom) Wiener processes. Dotted lines denote two standard deviations.

▶ 11.2 Spline Bayesian Quadrature for Smoother Integrands

If the prior behind the trapezoidal rule is too conservative, can we use the additional information, e.g. about the integrand's smoothness to build a better quadrature rule? More precisely, what if we know that the integrand is differentiable q times (but not what those derivatives are)? The Wiener process prior on f, which turned out to be the probabilistic analogue of the trapezoidal rule, is the simplest case of a more general class of quadrature rules based on polynomial spline interpolation. We can hence use the derivations provided in §5.4 to build new quadrature routines.

Let us assume that the integrand f is not just continuous, but has q continuous derivatives. We define the state

$$z(x) = \begin{bmatrix} F_a(x) \\ f(x) \\ f'(x) \\ \vdots \\ f^{(q)}(x) \end{bmatrix},$$

and again model $\mathrm{d}z(x) = Mz(x)\,\mathrm{d}x + L\,\mathrm{d}\omega_t$, now with the matrices M, L as defined in Eq. (5.22). For $q = 1$, we get the *integrated Wiener process*, for $q = 2$, the *twice-integrated Wiener process* (Figure 11.3) and so on. Posterior mean functions of the integrated Wiener process are cubic splines, those of the twice-integrated Wiener process are quintic splines, etc. (Figure 11.4). Using these two, and $H = [0, 1, 0, 0, \dots]$, the standard Kalman

1 **procedure** INTEGRATE($@f, a, b, N$)
2 $\quad x \leftarrow a$ // initialise state
3 $\quad \boldsymbol{m} \leftarrow [0; f(a); \mathbf{0}_q]$, $\tilde{P} \leftarrow$ e.g. Eq. (11.13) // define prior
4 $\quad A, \tilde{Q} \leftarrow$ e.g. Eqs. (5.25) & (5.26)
5 $\quad \alpha \leftarrow \alpha_0, \beta \leftarrow \beta_0$
6 $\quad$ **for** $i = 1, \dots, N-1$ **do**
7 $\qquad x \leftarrow x + \delta$ // move
8 $\qquad m^- = Am$ // predictive mean
9 $\qquad \tilde{P}^- = A\tilde{P}A^\intercal + \tilde{Q}$ // predictive covariance
10 $\qquad z = f(x) - Hm^-$ // observation residual
11 $\qquad s = H\tilde{P}^- H^\intercal$ // residual variance
12 $\qquad K = {}^1/_s \tilde{P}^- H^\intercal$ // gain
13 $\qquad m \leftarrow m^- + Kz$ // update mean
14 $\qquad \tilde{P} \leftarrow \tilde{P}^- - KsK^\intercal$ // update covariance
15 $\qquad \beta \leftarrow \beta + {}^{z^2}/_{2s}$ // update hyperparameter
16 $\quad$ **end for**
17 $\quad \mathbb{E}(F) \leftarrow m_1$ // point estimate
18 $\quad \operatorname{var}(F) \leftarrow \beta/(\alpha_0 + N/2 - 1) \cdot \tilde{P}_{11}$ // error estimate
19 $\quad r \leftarrow \beta - \beta_0 - \frac{N}{2}$ // model fit diagnostic (see §11.3)
20 $\quad$ **return** $\mathbb{E}(F), \operatorname{var}(F)$ // return mean, variance of integral
21 **end procedure**

Algorithm 11.2: General probabilistic quadrature rule in Kalman filter form, including marginalisation of the intensity θ^2. Lines 3–5 define the model, the rest is a generic form of the Kalman filter with conjugate prior hierarchical inference on θ^2. The notation $@f$ is meant to imply access to the function f itself at arbitrary inputs.

filter of Algorithm 5.1 turns into a probabilistic integration method. It is reproduced with the additional lines required for hyperparameter adaptation in Algorithm 11.2. The only terms requiring some care are the initialisation. Because we do not know – at least not without further computation – the derivatives of the integrand, we should be ignorant about them at initialisation. A simplistic way to encode this is to initialise $m = [0, f(a), 0, \dots]$ and

$$P = \begin{bmatrix} 0 & 0 & \mathbf{0}_{1\times q} \\ 0 & 0 & \mathbf{0}_{1\times q} \\ \mathbf{0}_{q\times 1} & \mathbf{0}_{q\times 1} & \alpha I_q \end{bmatrix}, \tag{11.13}$$

with a "very large" value of α.

Figure 11.5 shows empirical convergence of these rules when integrating our running example of Eq. (9.1), for the choices $q = 0$ (the trapezoidal rule) through $q = 3$.

▶ 11.3 Selecting the Right Model

We have seen how the performance of quadrature methods critically depends on how well their implicit assumptions – their model – describe the integrand. How should the model be

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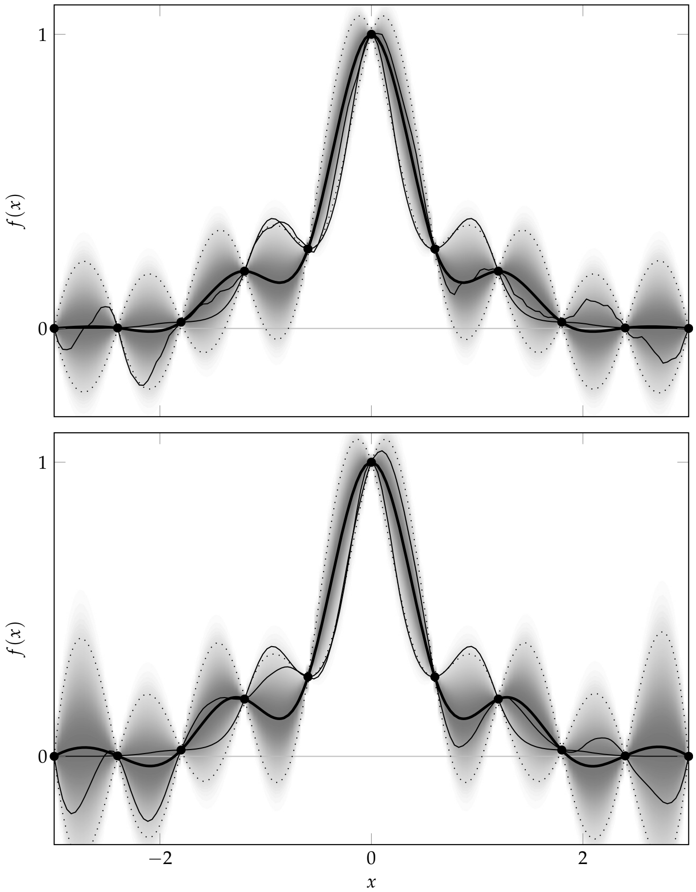

Figure 11.4: Posteriors over the integrand f of Eq. (9.1) arising from the integrated Wiener process (top, cubic spline mean) and twice-integrated Wiener process (bottom, quintic spline mean). Compare against the priors in Figure 11.3. For further explanation, see Figure 11.1.

selected? In the unusual case that the integrand is an element of the prior hypothesis space, performance can be increased by concentrating prior mass on the integrand. In the more realistic case that the integrand is *not* in the prior space, the issue changes to how much mass the prior assigns to good approximations that are *close* to the integrand.[16] In either case, designing a good quadrature rule for one specific problem involves finding a good combination of prior mean m and covariance k so that the three integrals of Eqs. (10.5), (10.6), and (10.7) are analytically available. Recall that we introduced a diverse range of tractable possibilities for Bayesian quadrature in §10.1. Exploring this space to find the best model is daunting.

[16] Kanagawa, Sriperumbudur, and Fukumizu (2020)

Even if we only consider Bayesian quadrature algorithms based on linear state-space models, there is a large space of potential SDEs to consider. In this setting, can the choice of model be automated? This search for the "right" prior model is itself another, potentially challenging, inference problem. In §11.1.4,

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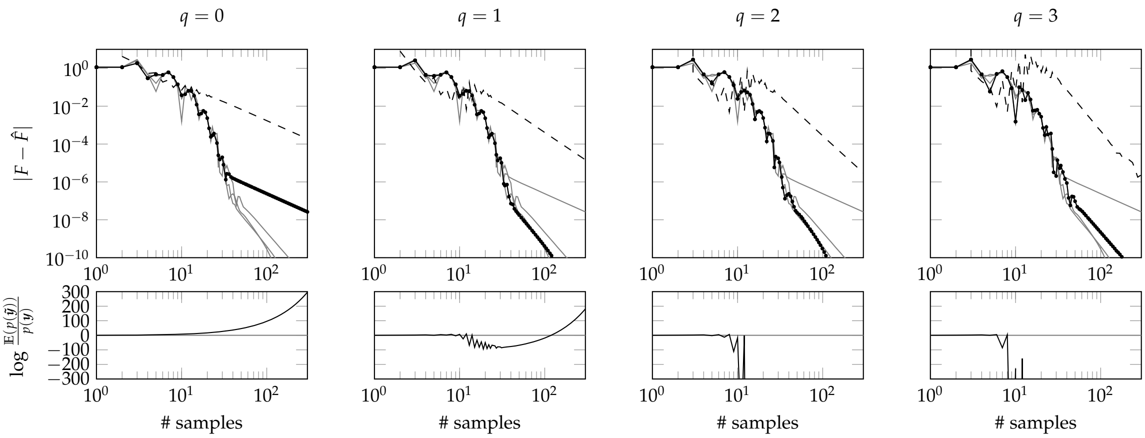

Figure 11.5: Convergence of quadrature rules from higher-order Wiener filters on the integrand of Eq. (9.1). The lower plots show the model fit statistic introduced in Eq. (11.15).

we tackled a simple version of this problem: we managed (that is, analytically marginalised, rather than selected) the model's parameter θ, the scale of uncertainty in the integral for the covariance family (10.9). For aspects of the prior other than this simple scalar term, such hierarchical inference is not always tractable. As such, its computational overhead is likely to be difficult to justify for a single integrand of modest evaluation cost.

However, inferring components of the model can still be interesting. Here we investigate doing so as a backstop mechanism, to detect if a numerical method is a poor match for the problem at hand. Consider the model class (10.9) with unit uncertainty-scale hyperparameter, $\theta = 1$. A likelihood for the model itself can be found by marginalising over the latent function f, as given in (6.13). The logarithm of this likelihood is a simple sum

$$\log p(Y \mid \mathcal{M}) = -\frac{1}{2}\sum_{i=1}^{N}\frac{(y_i - Hm_i^-)^2}{H\tilde{P}_i^- H^\intercal} - \frac{1}{2}\log|H\tilde{P}_i^- H^\intercal| + \text{const.} \tag{11.14}$$

To provide a reference baseline for this quantity, assume we have to *predict* this quantity before observing the values Y, but after having decided on the (regular) grid locations. Under the prior, the expected value of y_i is Hm_i^-, and the expected value of $(y_i - Hm_i^-)^2$ is just $H\tilde{P}_i^- H^\intercal$. Hence, the expected value of Eq. (11.14) for observables $\tilde{Y}$ drawn directly from the model is

$$\mathbb{E}_{\tilde{Y}|\mathcal{M}}\big(\log p(\tilde{Y} \mid \mathcal{M})\big) = -\frac{N}{2} - \frac{1}{2}\log|H\tilde{P}_i^- H^\intercal| + \text{const.}$$

The expected log-ratio between predicted and observed likeli-

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hood is

$$
\begin{aligned}
r(Y,\mathcal{M}) &:= \int \log \frac{p(\tilde{Y} \mid \mathcal{M})}{p(Y \mid \mathcal{M})} p(\tilde{Y} \mid \mathcal{M}) \,\mathrm{d}\tilde{Y} \\
&= -\frac{1}{2}\left(N - \sum_{i=1}^{N} \frac{(y_i - Hm_i^-)^2}{H\tilde{P}_i^- H^\intercal} \right) \qquad (11.15) \\
&= \sum_{i=1}^{N} \frac{z_i^2}{2s_i^2} - \frac{N}{2} = \beta_N - \beta_0 - \frac{N}{2},
\end{aligned}
$$

using the interior Kalman filter variables from Algorithm 11.2. We can see that the expected log-ratio is independent of the terms that only depend on design choices, rather than the data itself. It compares the variability of the observed data with its expectation. If $r(Y,\mathcal{M}) > 0$, then the integrand is less variable and more regular than expected under the prior, and we can expect the error estimate $\operatorname{var}_{\mathcal{M}}(F)$ to be conservative (i.e. the true error is likely smaller than the estimated one) – the numerical method is *under-confident*. On the other hand, if $r(Y,\mathcal{M}) < 0$, then the integrand varies more than the model would have predicted. Other, so far unexplored, areas are likely to lead to similar surprises, and thus the error estimate should not be trusted – it is *over-confident*.

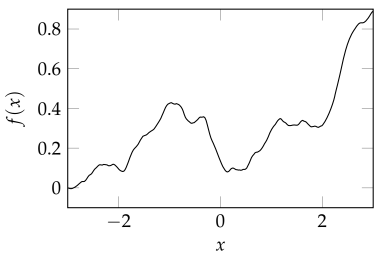

Figure 11.6: A draw from the integrated Wiener process. This particular function was used as the alternative integrand in the experiments represented in Figure 11.7.

Conveniently, the final line of Eq. (11.15) reveals that r is a trivial transformation of the sufficient statistic β, already collected in Algorithm 11.2! It can thus be computed at runtime without additional overhead. The bottom row of Figure 11.5 shows these values r during the integration process for the different models. We see that, although the integrand in this case is an infinitely smooth function, its derivative varies on a scale not expected by the higher-order Wiener processes. Based on the evidence, $q = 1$ appears like a good choice within this family of integrators – and indeed, the convergence rates do not rise further on increasing q.

Figure 11.7 shows another example: this time, the integrand is a true sample from an integrated ($q = 1$) Wiener process (the integrand is shown in Figure 11.6). As expected, the integrator based on the rough $q = 0$ process collects values for r that suggest its model assumptions are too rough, while the $q > 1$ integrators can detect their overconfident smoothness assumptions. For fixed values of θ, these integrators have over-confident error estimates. Posterior inference on θ adapts the error estimates so they become cautious again, but remain unreliable.

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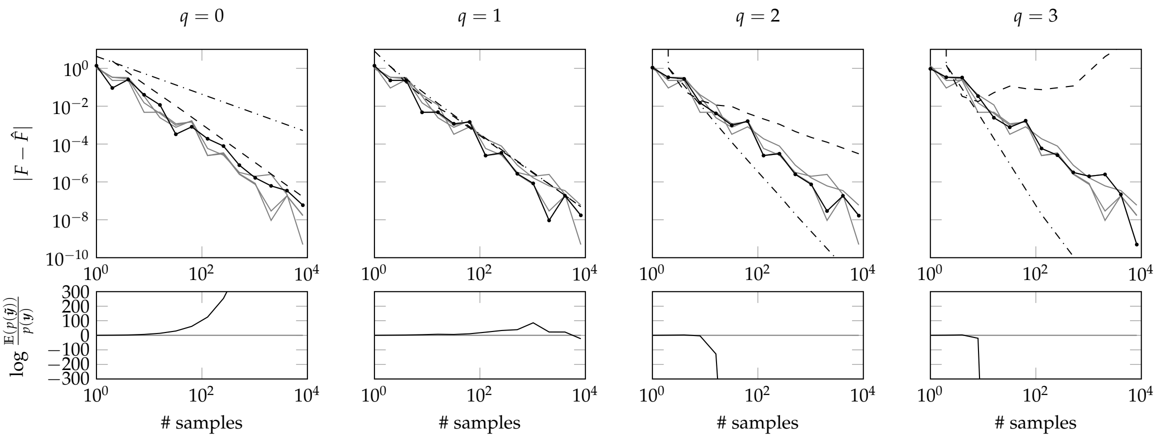

Figure 11.7: Convergence of quadrature rules when integrating a sample from a $q = 1$ times integrated Wiener process. Top row: error of the mean estimator (solid black, other values of q in grey for comparison). Dash-dotted lines are error estimates for a fixed value of θ, dashed lines are marginal variances under the Gamma prior on θ^{-2}. As in Figure 11.5, the bottom row shows the evolution of the model evidence statistic introduced in Eq. (11.15).

▶ 11.4 Connection to Gaussian Quadrature Rules

The detailed treatment above shows that the trapezoidal rule (both the regular grid design and the estimation rule), can be derived as a maximum a posteriori estimate arising from a Wiener process prior. The trapezoidal rule is a simple integration rule, often too simple for practical use. The most popular class of generic integration routines are known as *Gaussian quadrature rules*.[17] They are more advanced than the trapezoidal rule, and achieve significantly faster convergence on most practical problems. This section investigates the connection between Gaussian quadrature and Gaussian process regression, reviewing results by Karvonen and Särkkä (2017).

[17] The naming is due to a separate 1814 paper by Gauss, and unrelated to the Gaussian process prior used in Bayesian quadrature.

It will turn out that the probabilistic formulation of such methods is not particularly pleasing, because it yields a vanishing posterior variance on the estimate. These results nevertheless provide a reference point and intuition. The main reason to be interested in Gaussian and other classical quadrature rules, even from a probabilistic perspective, is that they are highly efficient. Figure 11.8 showcases the performance of the Gauss–Legendre on our running example, where its asymptotic convergence is very fast.

For this section, we denote the integration domain by $\Omega \subset \mathbb{R}$. Let ν be a probability measure on $\mathbb{R}$, and $f : \Omega \rightarrow \mathbb{R}$ denotes the integrand.[18] The task is to compute the integral

[18] Depending on the concrete quadrature rule in question, varying regularity requirements on Ω, ν, and f may apply. They will be assumed to hold without further comment.

$$\nu(f) = \int_\Omega f(x)\,\mathrm{d}\nu(x).$$

A *quadrature rule* $Q = (X, \boldsymbol{w})$ is a (linear) map of f to an approx-

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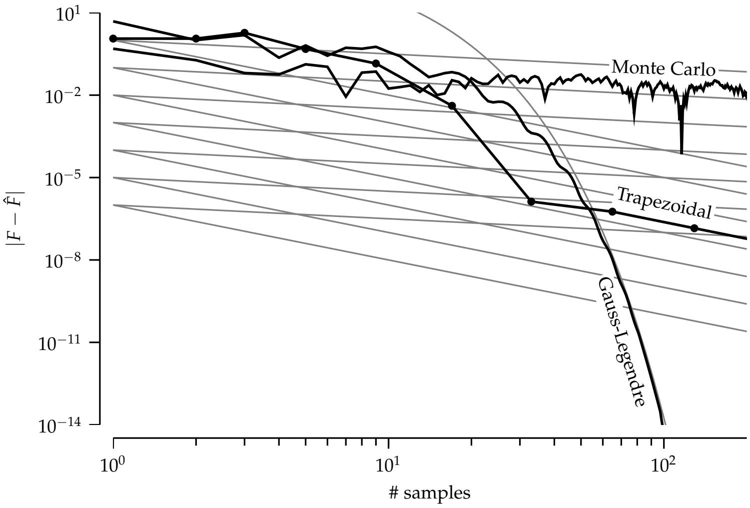

Figure 11.8: Error evolution for Gauss–Legendre quadrature on the running example of Eq. (9.1). In comparison to the Monte Carlo and Trapezoidal estimates introduced in previous sections, which each exhibit polynomial convergence rates as predicted by their corresponding error analysis, the Gaussian rule converges much faster. The curved grey line is a suggestive exponential function.

imation $Q(f)$ of $\nu(f)$ of the form

$$Q(f) := \sum_{i=1}^{N} w_i \cdot f(x_i),$$

where $X = [x_1, \ldots, x_N]$ are the *nodes* or *knots* (also sometimes called *sigma-points*) of the rule, and $\boldsymbol{w} = [w_1, \ldots, w_N]$ are the *weights*.

The most popular design criterion for quadrature rules is to require the rule to be *exact* for all polynomials up to some degree.

Definition 11.3. *Let $p_N(x) = \sum_{i=0}^{N} a_i x^i$ be a polynomial of degree N. A quadrature rule Q is of* degree *M if it is exact with respect to ν for* all *polynomials p_N of degree $N \leq M$, i.e. if*

$$Q(p_N(x)) = \nu(p_N(x)),$$

and inexact for at least one polynomial of degree $N = M + 1$.

For any set of N pairwise different nodes $\boldsymbol{x} \in \Omega^N$, there exists a quadrature rule[19] of degree at least $N - 1$. But the punchline of classical quadrature is that, if the nodes are chosen carefully, it is in fact possible to achieve degree $2N - 1$ with those N integrand evaluations. Such rules are called *Gaussian quadrature rules*:[20]

Theorem 11.4 (Gaussian quadrature)**.** *For sufficiently regular ν, there exists a unique quadrature rule with N nodes of degree $2N - 1$. Its nodes are given by the roots of the Nth ν-orthogonal polynomial[21] ψ_N, and its weights $\boldsymbol{w}$ are all positive.*

[19] That rule is identified by the weights $\boldsymbol{w}$ satisfying the N monomial constraints $Q(x^i) = \nu(i)$ for $i = 0, \ldots, N-1$, which amount to the Vandermonde system

$$\begin{bmatrix} x_1^0 & \cdots & x_N^0 \\ \vdots & \ddots & \vdots \\ x_1^{N-1} & \cdots & x_N^{N-1} \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_N \end{bmatrix} = \begin{bmatrix} \nu(x^0) \\ \vdots \\ \nu(x^{N-1}) \end{bmatrix}.$$

[20] Theorem 11.4 is a simplified form of a more general result. For more see, e.g., p. 97 of Davis and Rabinowitz (1984). A proof is in Thms. 1.46–1.48 in §1.4 of Gautschi (2004).

[21] A sequence of polynomials $\{\psi_0(x), \psi_1(x), \ldots\}$ (where ψ_i is of degree i) are called *orthogonal* under the measure ν if

$$\nu(\psi_i \cdot \psi_j) = \int \psi_i(x) \cdot \psi_j(x) \, d\nu(x) = c_i \delta_{ij}.$$

For each sufficiently regular ν, this sequence exists and is uniquely identified up to the constants c_i (see §1.4 in Gautschi (2004)).

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Ω	$\nu(x)$	$\{\psi_i\}$
$[-1,1]$	$1/2$	Legendre
$[-1,1]$	$1/\sqrt{1-x^2}$	Chebyshev
$[-1,1]$	$(1-x)^\alpha(1+x)^\beta$	Jacobi
$[0,\infty)$	e^{-x}	Laguerre
$(-\infty,\infty)$	e^{-x^2}	Hermite

Table 11.1: Popular cases of integration domain Ω, (unnormalised) measure ν and corresponding set of orthonormal polynomials ψ. The parameters of the measure for the Jacobi polynomials are two reals $\alpha, \beta > -1$. The special case $\alpha = \beta$ gives rise to what are termed Gegenbauer polynomials and Gauss–Gegenbauer quadrature.

A relatively small dictionary of domains, base measures and corresponding orthogonal polynomials have found wide use in practice. Some of them are listed in Table 11.1.

Karvonen and Särkkä (2017) provided the connection between these foundational rules and Bayesian quadrature. Let $[\bar{\psi}_i]_{i=0,1\dots}$ be a set of ortho*normal* polynomials. That is,

$$\nu(\bar{\psi}_i\bar{\psi}_j) = \int \bar{\psi}_i(x)\bar{\psi}_j(x)\,\mathrm{d}\nu(x) = \delta_{ij}.$$

Now consider the Gaussian process $p(f) = \mathcal{GP}(f;0,k)$ arising from the degenerate (i.e. finite-rank) kernel

$$k^q(x,x') = \sum_{i=0}^{q-1} c_i\bar{\psi}_i(x)\bar{\psi}_i(x'), \tag{11.16}$$

and assume a set of positive scales $[c_i]_{i=0,1,\dots} \subset \mathbb{R}_+$. We might want to call this a *polynomial kernel*, but that name is already overloaded in the Gaussian process literature.

Recall from §4.1 that taking the degenerate kernel (11.16) amounts to the assumption that f can be written as $f(x) = \sum_i \bar{\psi}_i(x)v_i$ with weights $v_i \in \mathbb{R}$ drawn i.i.d. from $p(v_i) = \mathcal{N}(v_i;0,c_i)$. The following theorem shows that, for any design $\boldsymbol{x}$, the Bayesian quadrature algorithm arising from the Gaussian process prior $p(f) = \mathcal{GP}(f;0,k^q)$ yields a quadrature rule of non-trivial degree, and the design choice that is optimal under the Bayesian perspective coincides with the Gaussian quadrature rule (i.e. the optimal one under the classic interpretation).

Theorem 11.5 (Karvonen and Särkkä, 2017). *The Bayesian quadrature rule with the kernel k^q on (Ω,ν) coincides with the classical quadrature rule Q of degree $M-1$ if, and only if, $N \leq q \leq M$. For such a rule, the posterior variance on the integral vanishes: $\mathfrak{v} = 0$.*

After setting $N := \frac{q}{2}$, we obtain the following statement.

Corollary 11.6 (Bayesian Gaussian Quadrature). *For each $N \in \mathbb{N}$, there is a unique N-point optimal Bayesian quadrature rule for the kernel k^{2N} on (Ω,ν), and it coincides with the Gaussian quadrature rule.*

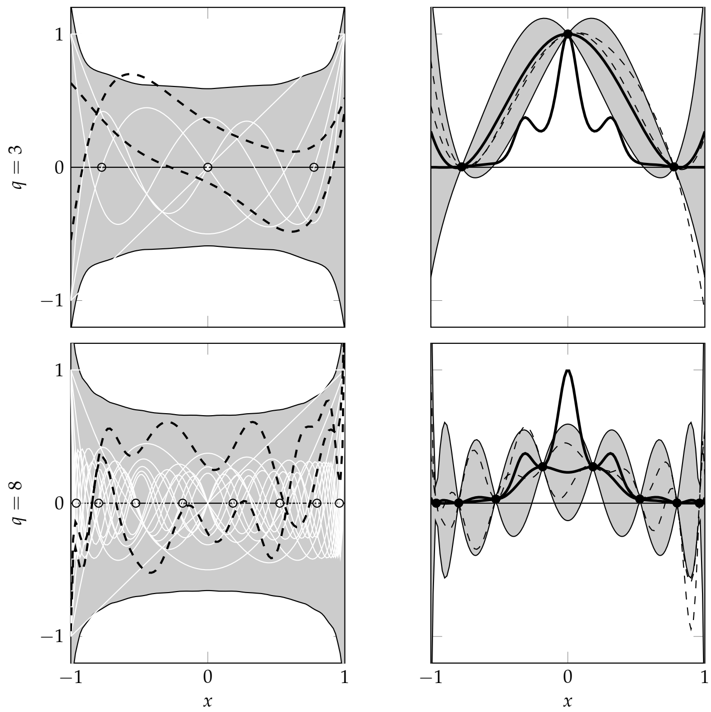

Figure 11.9: Probabilistic interpretation of Gauss–Legendre integration (that is, for $\nu(x)$ the Lebesgue measure). Priors (left) and posteriors (right) consistent with the Gauss–Legendre rules of degree $2q - 1 = 5$ (top) and 15 (bottom), respectively. The left plots show two marginal standard deviations as a shaded area, the first $2q - 1$ Legendre polynomials spanning the kernel (which are exactly integrated by the associated quadrature rule) in white, and two samples from the prior in dashed black. The right panels show the posterior after q evaluations at the nodes of the qth polynomial, again with two standard deviations as a shaded region, the posterior mean and the integrand in thick black, and two samples in dashed thin black. Even though the posterior variance on the integrand f is non-zero, the posterior variance on the integral F vanishes – the two shown samples have the same integral as the posterior mean. Changes in the scales c_i of the individual polynomials can change the means, marginal variances, and samples shown in these figures drastically, while leaving the resulting integral estimates identical.

Although this result shows certain Bayesian quadrature rules to be equivalent to Gaussian quadrature, it also highlights the *differences* between the probabilistic and the classic approach. Three key aspects are particularly worth noting:

- The equivalence class of GP priors whose associated BQ rules coincide with the Gaussian rule is large, because it includes all choices of $\boldsymbol{c} \in \mathbb{R}_+^{2N}$ in the prior

$$p(\boldsymbol{v}) = \mathcal{N}\big(\boldsymbol{v}; 0, \operatorname{diag}(\boldsymbol{c})\big)$$

  for the weights $\boldsymbol{v}$ of $f(x) = \sum_{i=0}^{2N} v_i \bar{\psi}_i(x)$. Figure 11.9 shows priors for the particular choice $\boldsymbol{c} = \mathbf{1}$, but changes to these weights can give rise to wildly different priors in the space of f. If the variances c_i decay rapidly with growing i, the prior assumes a more regular integrand, while a growing sequence of c_i amounts to the assumption of a "more variant" (though always smooth) integrand. All these choices, though, yield the same integral estimate when used in the setting of Corollary 11.6, i.e. when there are N observations at the nodes of the Nth polynomial.

- Under the posterior on f arising from the kernel k^{2N}, the variance is zero at the N nodes X of the Nth polynomial, but is generally non-zero at $x \notin X$ (see Figure 11.9). To explain this

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situation, note that in a set of ν-orthogonal polynomials with $c_0 > 0$, all but the constant one have vanishing expectation under ν. That is, they satisfy

$$\nu(\psi_i) = 0, \ \forall i > 0,$$

because $\psi_i(x) = \pm\psi_i(x)\psi_0(x)/\sqrt{c_0}$. In the setting of Corollary 11.6, those N evaluations exactly identify the value of the first coefficient v_0, but not necessarily those of the other coefficients. So there is flexibility left in the function values, but only in ways that do not contribute to the integral. In the posteriors shown in Figure 11.9, all sampled hypotheses, and the posterior means, share the same integral.

⊖ A related feature is that the prior of Eq. (11.16) must be parametric (of finite rank, degenerate) to yield Gaussian quadrature rules. But of course, the number N of evaluations is meant to grow as the algorithm runs. From the probabilistic perspective, this means the prior constantly becomes more general, more flexible, as N grows. Of course, its dependence on data, through N, means that this prior is not a real prior at all, at least one that an orthodox Bayesian would recognise. The price to be paid for this oddity is that it is difficult to associate a meaningful notion of uncertainty with the posterior in this simple form, because $\mathfrak{v} = 0$. It remains an open question at the time of writing, however, whether an empirical Bayesian extension is possible.

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